

SM PHARMACEUTICALS SDN. BHD.

CLARITROX (CLARITHROMYCIN) 250 AND 500 TABLET

DESCRIPTION:

250mg: A yellow colour, modified capsule shaped, SM embossed with break-line on one side, film coated tablet.

500mg: A yellow colour, modified capsule shaped, with break-line on one side, film coated tablet.

COMPOSITION:

Each film-coated tablet contains: Clarithromycin 250 mg or 500 mg.

ACTIONS AND MODE OR MECHANISMS OF ACTION:

Actions: Clarithromycin is a semisynthetic macrolide antibiotic.

Mechanism of action:

Clarithromycin binds to the 50S ribosomal subunit of the 70S ribosome of susceptible organisms, thereby inhibiting bacterial RNA-dependent protein synthesis.

Microbiology:

Clarithromycin is active *in vitro* against a variety of aerobic and anaerobic gram positive and gram-negative microorganisms.

Additionally, the 14-OH clarithromycin metabolite also has clinically significant antimicrobial activity. The 14-OH clarithromycin is twice as active against *Haemophilus influenzae* microorganisms as the parent compound.

Clarithromycin has been shown to be active against most strains of the following microorganisms both *in vitro* and in clinical infections.

Aerobic Gram-positive microorganisms - *Staphylococcus aureus*, *Streptococcus pneumoniae* and *Streptococcus pyogenes*

Aerobic Gram-negative microorganisms - *Haemophilus influenzae* and *Moraxella catarrhalis*

Other microorganisms - *Mycoplasma pneumoniae* and *Chlamydia pneumoniae* (TWAR)

Mycobacteria

Mycobacterium avium complex (MAC) consisting of: *Mycobacterium avium*, *Mycobacterium intracellulare* and *Helicobacter pylori*

Clarithromycin has been shown to be active against most strains of *Helicobacter pylori* *in vitro* and in clinical infections when combined with omeprazole, lansoprazole and amoxicillin, or ranitidine bismuth citrate.

The following *in vitro* data are available, but their clinical significance is unknown.

Aerobic Gram-positive microorganisms - *Streptococcus agalactiae*, *Streptococci* (Groups C,F,G) and Viridans group streptococci

Aerobic Gram-negative microorganisms - *Bordetella pertussis*, *Legionella pneumophila* and *Pasteurella multocida*

Anaerobic Gram-positive microorganisms - *Clostridium perfringens*, *Peptococcus niger* and *Propionibacterium acnes*

Anaerobic Gram-negative microorganisms - *Prevotella melaninogenica* (formerly *Bacteroides melaninogenicus*)

PHARMACOLOGY (SUMMARY OF PHARMACODYNAMICS AND PHARMACOKINETICS):

Absorption:

Well absorbed from the gastrointestinal tract; stable in gastric acid; food delays the rate, but not the extent, of absorption; bioavailability is approximately 55% in healthy volunteers.

Distribution and Elimination:

Widely distributed into tissue and fluids, high concentrations found in nasal mucosa, tonsils, and lungs; concentrations in tissue are higher than those in serum because of high intracellular concentrations.

Binding to Plasma Proteins: 65 to 75%.

Biotransformation: Hepatically metabolised.

Half - life:

Normal renal function - 250 mg every 12 hours - 3 to 4 hours.

- 500 mg every 12 hours - 5 to 7 hours.

Renal function impairment (creatinine clearance of < 30 ml per minute) -

Approximately 22 hours.

Peak serum concentration: Clarithromycin: Steady-state:

250 mg every 12 hours: Approximately 1 mcg/mL

500 mg every 12 hours: 2 to 3 mcg/mL

14 - Hydroxyclearithromycin: Steady- state:

250 mg every 12 hours: Approximately 0.6 mcg/mL.

500 mg every 12 hours: Up to 1 mcg/mL.

Elimination: Renal - Approximately 20 and 30 %, respectively, of the dose of 250- and 500-mg tablets given twice a day is excreted in the urine as unchanged drug. Faecal - Approximately 4 % of a 250-mg dose is excreted in the faeces

INDICATIONS:

Treatment of infections due to susceptible organisms of the upper respiratory tract infections (e.g. streptococcal pharyngitis/tonsillitis, acute maxillary sinusitis), lower respiratory tract infections (e.g. bronchitis, pneumonia), acute otitis media, skin and skin structure infections (e.g. impetigo, folliculitis, cellulitis, abscesses).

CONTRAINDICATIONS:

Clarithromycin is contraindicated in patients with a known hypersensitivity to clarithromycin, erythromycin, or any of the macrolide antibiotics.

Concomitant administration of clarithromycin with cisapride, pimozide, or terfenadine is contraindicated. There have been post-marketing reports of drug interactions when clarithromycin and/or erythromycin are co-administered with cisapride, pimozide, or terfenadine resulting in cardiac arrhythmias (QT prolongation, ventricular tachycardia, ventricular fibrillation, and torsades de pointes) most likely due to inhibition of hepatic metabolism of these drugs by erythromycin and clarithromycin. Fatalities have been reported. Similar effects have been observed with concomitant administration of astemizole and other macrolides. Clarithromycin and ergot derivatives should not be co-administered.

Concomitant administration of Clarithromycin and the following drugs is contraindicated: Domperidone as this may results in QT prolongation and cardiac arrhythmias including ventricular tachycardia, ventricular fibrillation, and torsades de pointes (See Section Interactions).

SIDE EFFECTS / ADVERSE REACTIONS:

The most frequently reported events in adults were diarrhoea, nausea, abnormal taste, dyspepsia, abdominal pain/discomfort, and headache. Most of these events were described as mild or moderate in severity. Of the reported events, only 1 % was described as severe. Allergic reactions ranging from urticaria and mild skin eruptions to rare cases of anaphylaxis and Steven Johnson's syndrome have occurred. Other spontaneously reported events include glossitis, stomatitis, oral moniliasis, vomiting, tongue discoloration, and dizziness. Transient CNS events including anxiety, behavioural changes, confusional states, depersonalisation, disorientation, hallucinations, insomnia, nightmares, psychosis, tinnitus and vertigo have been reported. Hepatic dysfunction, including increased liver enzymes and hepatocellular and /or cholestatic hepatitis, with or without jaundice, has been infrequently reported. Rarely, clarithromycin have been associated with ventricular arrhythmias, including ventricular tachycardia and torsades de pointes, in individuals with prolonged QT intervals. Pseudomembranous colitis has been reported rarely with clarithromycin, and may range in severity from mild to life threatening. There have been report of pancreatitis and convulsion.

Skin and Subcutaneous Tissue Disorders

Frequency not known severe cutaneous adverse reactions (SCARs) including Steven-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) and acute generalised exanthematous pustulosis (AGEP).

PRECAUTIONS / WARNINGS:

Cross sensitivity and/or related problems

Patients who are hypersensitive to erythromycin or other macrolides may also be hypersensitive to clarithromycin.

Pregnancy:

There are no adequate and well-controlled studies in pregnant women. Clarithromycin should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus. (See WARNINGS.)

Breast Feeding

Clarithromycin and its active metabolite are distributed into human breast milk. Therefore, caution should be exercised when clarithromycin is administered to a nursing woman.

Hepatic and renal function impairment

Clarithromycin is principally excreted by the liver. Therefore, caution should be exercised in administering the antibiotic to patients with impaired hepatic function. Caution should also be exercised when administering clarithromycin to patients with moderate to severe renal failure.

WARNINGS

CLARITHROMYCIN SHOULD NOT BE USED IN PREGNANT WOMEN EXCEPT IN CLINICAL CIRCUMSTANCES WHERE NO ALTERNATIVE THERAPY IS APPROPRIATE. IF PREGNANCY OCCURS WHILE TAKING THIS DRUG, THE PATIENT SHOULD BE APPRISED OF THE POTENTIAL HAZARD TO THE FETUS. CLARITHROMYCIN HAS DEMONSTRATED ADVERSE EFFECTS OF PREGNANCY OUTCOME AND/OR EMBRYO-FETAL DEVELOPMENT IN MONKEYS, RATS, MICE, AND RABBITS AT DOSES THAT PRODUCED PLASMA LEVELS 2 TO 17 TIMES THE SERUM LEVELS ACHIEVED IN HUMANS TREATED AT THE MAXIMUM RECOMMENDED HUMAN DOSES. (See PRECAUTIONS - *Pregnancy*)

Pseudomembranous colitis has been reported with nearly all anti-bacterial agents, including clarithromycin, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhea subsequent to the administration of anti-bacterial agents. Prolonged or repeated use of clarithromycin may result in an overgrowth of non-susceptible bacterial or fungi. If super infection occurs, clarithromycin should be discontinued and appropriate therapy instituted.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g., Steven-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) and acute generalised exanthematous pustulosis (AGEP), Claritrox Tablet should be discontinued immediately and appropriate treatment should be urgently initiated.

DRUG INTERACTIONS:

Carbamazepine - (administration of carbamazepine with clarithromycin has been shown to increase significantly the plasma concentration of carbamazepine; carbamazepine serum levels should be monitored)

Digoxin - (concurrent administration of digoxin with clarithromycin has been shown to increase serum digoxin concentrations; monitoring of digoxin serum levels is recommended)

Rifabutin or Rifampin - (concurrent use of clarithromycin and rifabutin or rifampin decreases the serum concentration of clarithromycin by > 50 %)

Terfenadine - (concurrent use of clarithromycin and terfenadine may increase the plasma concentration of terfenadine and its active metabolite by 2 to 3 times; concurrent use should be avoided)

Theophylline - (concurrent administration with clarithromycin has been shown to increase the AUC of theophylline by 17 %; monitoring of theophylline serum levels is recommended)

Warfarin - (concurrent administration with clarithromycin has been shown to potentiate the effects of warfarin; prothrombin time should be closely monitored)

Ergotamine or dihydroergotamine - (concurrent use with clarithromycin has been associated in some patients with acute ergot toxicity)

Triazolam - (There have been reports of CNS effects with the concomitant use of clarithromycin and triazolam)

Cyclosporine, tacrolimus, hexobarbital, phenytoin, alfentanil, Quinidine/disopyramide, lovastatin, bromocriptine, valproate, astemizole and other drugs metabolised by the cytochrome P450 system - (concurrent use with clarithromycin may be associated with elevations in serum levels of these other drugs; serum concentrations should be closely monitored)

Drug metabolised by CYP3A isoenzyme - Clarithromycin has the potential to interact with the large number of drugs through its action on hepatic cytochrome P450 isoenzymes, particularly CYP1A2 and CYP3A4. Macrolides inhibit drug metabolism by microsomal cytochromes by competitive inhibition and by the formation of inactive complexes. Such interactions can result in severe adverse effects, including ventricular arrhythmias with the non-sedative antihistamines astemizole and terfenadine and with the prokinetic drug cisapride. Clarithromycin is less likely to inhibit the hepatic metabolism of other drugs, although those undergoing first-pass metabolism may still be affected.

Zidovudine - Simultaneous oral administration of zidovudine and clarithromycin resulted in a lower peak serum concentration [C_{max}], lower AUC, and delayed time to peak concentration [T_{max}] of zidovudine should be taken at least hours apart.

Maalox and Ranitidine - Increased plasma concentrations of clarithromycin may also occur when it is co-administered with Maalox or ranitidine. No adjustment to the dosage is necessary.

Omeprazole - In a study in healthy subjects, increased concentrations of clarithromycin and its active metabolite were observed in gastric tissue and mucus and to lesser extent in plasma during concomitant administration of omeprazole. In addition, administration of clarithromycin with omeprazole resulted in higher and more prolonged plasma concentrations of omeprazole.

Co-administration of Clarithromycin, known to inhibit CYP3A, and a drug primary metabolized by CYP3A may be associated with elevations in drug concentrations that could increase or prolong both therapeutic and adverse effect of the concomitant drug. The following drugs or drug classes are known or suspected to be metabolized by CYP3A isozyme: Domperidone.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:

Clarithromycin tablets may be given with or without food.

Adults: 250 mg every 12 hours for 7 days, increased in severe infections to 500 mg every 12 hours for up to 14 days. In patients with renal impairment with creatinine clearance < 30 ml/min, the dosage should be reduced by half. Dosage should not be continued beyond 14 days in these patients.

SYMPTOMS AND TREATMENT FOR OVERDOSE AND ANTIDOTES:

Reports indicate that the ingestion of large amounts of clarithromycin can be expected to produce gastrointestinal symptoms. One patient who had a history of bipolar disorder ingested 8 g of clarithromycin and showed altered mental status, paranoid behaviour, hypokalemia and hypoxemia. Allergic reactions accompanying overdosage should be treated by the prompt elimination of unabsorbed drug and supportive measures.

PACKING / PACK SIZES:

Blisters of 4's/Box and 10 x 4's/Box, Blisters of 10's/Box and 10 x 10's/Box. Blister of 7's/Box and 14's/Box, 3x10's/Box and 2x10's (for 500 only).

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

Store below 30°C. Protect from light.

Auxiliary labelling

* Continue medicine for full time of treatment

SHELF LIFE: 3 years

MANUFACTURER AND PRODUCT REGISTRATION HOLDER:

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